

# HAT-P-36b

R-0779

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-36\_250523 · created 2026-07-28T06:00:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460818.8178 +/- 0.0044 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.143 +/- 0.017                 |
| Transit depth (Rp/Rs)^2                           | 2.05 +/- 0.49 [%]               |
| Orbital Inclination (inc)                         | 86.99 +/- 2.53                  |
| Airmass coefficient 1 (a1)                        | 1.025 +/- 0.018                 |
| Airmass coefficient 2 (a2)                        | -0.0127 +/- 0.0072              |
| Scatter in the residuals of the lightcurve fit is | 1.79 %                          |
| Best Comparison Star                              | #2 - [160, 423]                 |
| Optimal Aperture                                  | 3.29                            |
| Optimal Annulus                                   | 10.43                           |
| Transit Duration (day)                            | 0.0958 +/- 0.0045               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                      |
|--------------------------------------|------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.143 $\pm$ 0.017 vs 0.126 (deviation 1.0 $\sigma$ ) |
| Duration fit vs published            | 0.0958 d vs 0.0925 d (deviation 0.73 $\sigma$ )      |
| Mid-time O–C                         | -4.1 min                                             |
| Depth SNR                            | 8.4                                                  |
| Residual scatter                     | 1.79 %                                               |

## Light curve

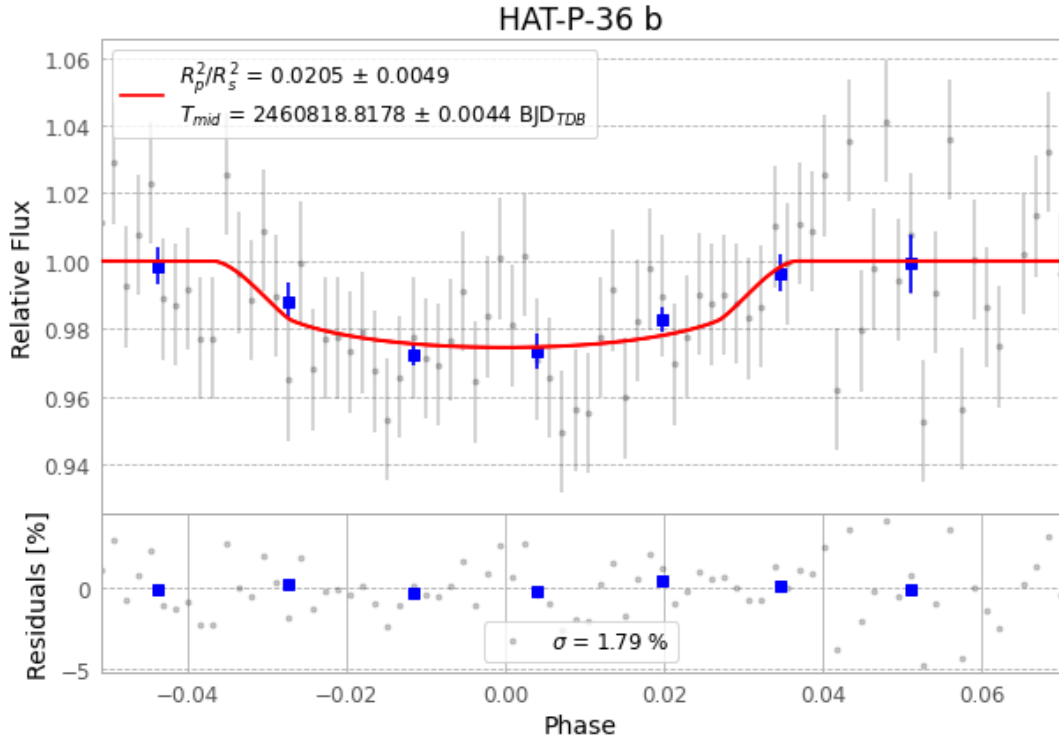


Figure 1 — FinalLightCurve\_HAT-P-36 b\_2025-05-22.png (SHA-256 83b7ae2484f5e0da...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 276], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 457c4bdf95c2c84b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

78 FITS frames (51 MB), HATP-36250523055715.FITS → HATP-36250523094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250523.log (f9e6809eff28...), FinalLightCurve\_HAT-P-36 b\_2025-05-22.png (83b7ae2484f5...), FinalLightCurve\_HAT-P-36 b\_2025-05-22.csv (beefe7907604...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 06:00Z.